Strict separable non-saturation for a seven-state prepare-and-measure witness

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Abstract

Absolutely entangled sets can be certified in a dimension-bounded prepare-and-measure experiment when a correlation witness exceeds its maximum over separable preparations. For a recently proposed seven-state two-qubit ensemble, the target value is $3/4$, whereas alternating optimization found a separable value 0.74990679 and no relaxation was known to place the separable maximum strictly below $3/4$. We prove strict non-saturation for this fixed witness. Equality in the Helstrom, frame-purity, and Cauchy–Schwarz bounds would require seven pure product rays to reproduce all 21 transition probabilities of the target ensemble. The six zero transitions admit 729 assignments to the two local qubit factors. Exact orthogonality closure leaves nine classes up to exchange of the factors. Eight have unit Gröbner ideal over $\mathbb{Q}(\sqrt{7})$; the remaining class forces a local Bloch Gram matrix with determinant $-27/16$ and is therefore unphysical. Compactness then gives $Q_{\mathrm{SEP}} < 3/4$. The proof establishes existence of a positive separable gap, but it neither computes the optimum nor provides a numerical robustness threshold.

Keywords: absolutely entangled set; prepare-and-measure witness; separable bound; product rays; exact algebraic computation

1 Introduction

Entanglement depends on a tensor-product decomposition of the Hilbert space. An absolutely entangled set (AES) is a collection of states such that, for every global change of basis, at least one member of the transformed set is entangled. Explicit minimal examples in two-qubit systems were introduced by Cai *et al.* [1]. Prepare-and-measure experiments offer an operational, semi-device-independent route to this set property: the communicated Hilbert space has a known dimension, while the preparation and measurement devices are otherwise uncharacterized. Prepare-and-measure self-testing provides the general setting for witnesses built from pairwise state distinguishability [2].

Patra, Taoutioui, and Vértesi recently constructed such witnesses for extended AES ensembles [3]. Their seven-state example $\widetilde{S}_3$ has unrestricted value $3/4$. A see-saw search found the feasible separable value 0.74990679, but the available semidefinite relaxation did not produce an upper bound below $3/4$. Thus a separable strategy saturating the unrestricted value was not excluded.

Here we settle this equality question for the exact seven-state instance. We first derive a necessary equality condition that does not assume a Wigner property: saturation would require seven product rays having all the target transition probabilities. We then exclude such rays by an exhaustive exact calculation followed by one elementary Bloch-Gram obstruction. The resulting statement is deliberately qualitative,

$$Q_{\mathrm{SEP}} < \frac{3}{4}. \tag{1}$$

It proves that a gap exists, but supplies neither its magnitude nor the value of Q_{SEP}. Consequently, it does not yet give a finite-precision experimental threshold.

2 The fixed witness

Let $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$, with basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$ and $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. The target rays are

$$\begin{aligned} |\psi_1\rangle &= |00\rangle, & |\psi_2\rangle &= |0+\rangle, \\ |\psi_3\rangle &= |+0\rangle, & |\psi_4\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \end{aligned} \tag{2}$$

$$\begin{aligned} |\psi_5\rangle &= \frac{|01\rangle}{\sqrt{6}} - \sqrt{\frac{2}{3}}|10\rangle + \frac{|11\rangle}{\sqrt{6}}, \\ |\psi_6\rangle &= |-\rangle \otimes |1\rangle, \\ |\psi_7\rangle &= a_1|00\rangle - b_1(|01\rangle + |10\rangle + |11\rangle). \end{aligned} \tag{3}$$

where

$$a_1 = \sqrt{\frac{7-2\sqrt{7}}{14}}, \qquad b_1 = \sqrt{\frac{7+2\sqrt{7}}{42}}. \tag{4}$$

Writing $P_i = |\psi_i\rangle\langle\psi_i|$, the positive weights are

$$\alpha_1 = \cdots = \alpha_4 = \frac{1}{6+2\sqrt{7}}, \qquad \alpha_5 = \alpha_6 = \frac{\sqrt{7}-1}{8}, \qquad \alpha_7 = \frac{3\sqrt{7}-7}{4}. \tag{5}$$

Direct calculation gives the tight-frame identities

$$\sum_{i=1}^{7} \alpha_i = 1, \qquad \sum_{i=1}^{7} \alpha_i P_i = \frac{I_4}{4}. \tag{6}$$

For every unordered pair $1 \le j < i \le 7$, Bob has an independent binary effect $0 \preceq E_{ij} \preceq I_4$. With

$$c_{ij} = \alpha_i \alpha_j \|P_i - P_j\|_1, \tag{7}$$

the witness is

$$W(\rho, E) = \sum_{j<i} c_{ij} \operatorname{Tr}[(\rho_i - \rho_j)E_{ij}]. \tag{8}$$

The separable benchmark Q_{SEP} is the attained maximum of (8) over seven trace-one states $\rho_i \in \mathrm{SEP}(2{:}2)$ and the 21 effects. This is a finite, dimensionless, single-round optimization. No common measurement constraint is imposed between distinct pairs.

Set

$$t_{ij} = \operatorname{Tr}(P_i P_j) = |\langle\psi_i|\psi_j\rangle|^2. \tag{9}$$

For the target preparations, each effect may be chosen as the Helstrom projector for $P_i - P_j$. Its contribution is $2\alpha_i\alpha_j(1 - t_{ij})$. Using (6),

$$2\sum_{j<i} \alpha_i \alpha_j (1 - t_{ij}) = 1 - \operatorname{Tr}\left(\sum_i \alpha_i P_i\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}. \tag{10}$$

This reproduces the unrestricted target value.

3 What equality would require

Proposition 1. If a separable strategy attains $W = 3/4$, then there are seven pure product projectors R_i satisfying

$$\mathrm{Tr}(R_i R_j) = t_{ij} \qquad (1 \le j < i \le 7). \tag{11}$$

Proof. The feasible preparation and effect sets are compact, so a maximizing strategy exists. With the effects fixed, the score is linear in each preparation. Every separable preparation is a convex combination of pure product projectors; replacing it by a component that maximizes the relevant linear functional cannot lower the score. Repeating this for all seven preparations shows that some optimum has pure product preparations R_i.

Put $s_{ij} = \mathrm{Tr}(R_i R_j)$ and $F = \sum_i \alpha_i R_i$. Optimizing each binary effect independently gives

$$W \le 2 \sum_{j<i} \alpha_i \alpha_j \sqrt{(1 - t_{ij})(1 - s_{ij})}. \tag{12}$$

The two squared norms appearing in Cauchy–Schwarz are

$$A = \sum_{j<i} \alpha_i \alpha_j (1 - t_{ij}) = \frac{3}{8}, \tag{13}$$

$$B = \sum_{j<i} \alpha_i \alpha_j (1 - s_{ij}) = \frac{1 - \mathrm{Tr}\, F^2}{2} \le \frac{3}{8}. \tag{14}$$

Here (13) follows from (10); in (14), F is a density operator on a four-dimensional space, so $\mathrm{Tr}\, F^2 \ge 1/4$. Therefore

$$W \le 2\sqrt{AB} \le \frac{3}{4}. \tag{15}$$

If $W = 3/4$, every inequality in (12)–(15) is an equality. Hence $A = B$, and equality in Cauchy–Schwarz makes the two 21-component vectors

$$\big(\sqrt{\alpha_i \alpha_j (1 - t_{ij})}\big)_{j<i}, \qquad \big(\sqrt{\alpha_i \alpha_j (1 - s_{ij})}\big)_{j<i} \tag{16}$$

proportional. Their norms agree. All weights are positive and all target rays are distinct, so the proportionality factor is one and $s_{ij} = t_{ij}$ for every pair. This proves (11). □

The proposition only uses equality in metric and purity bounds. In particular, it does not require the transition probabilities to determine the target rays up to a unitary or antiunitary, and hence does not invoke a Wigner property.

4 Exact product-ray obstruction

Write a product ray as $R_i = |a_i\rangle\langle a_i| \otimes |b_i\rangle\langle b_i|$. For unit Bloch vectors $\boldsymbol{r}_i^A, \boldsymbol{r}_i^B \in S^2$, define the local fidelities

$$q_{ij}^X = |\langle x_i | x_j \rangle|^2 = \frac{1 + \boldsymbol{r}_i^X \cdot \boldsymbol{r}_j^X}{2}, \qquad X \in \{A, B\}. \tag{17}$$

Condition (11) becomes

$$q_{ij}^A q_{ij}^B = t_{ij}. \tag{18}$$

The complete target matrix is given in Appendix A. Its zero graph is

$$Z = \{15, 16, 36, 56, 57, 67\}, \tag{19}$$

where, for example, 15 denotes the pair $\{1, 5\}$.

We use the following finite exact test. Each edge in Z is assigned to factor A, factor B, or both, according to which local fidelity vanishes. This gives $3^6 = 729$ assignments. For a qubit

ray, the orthogonal ray is unique: in Bloch coordinates an orthogonality edge makes the two vectors antipodal. Thus every connected local zero graph has a parity labeling; equal-parity vertices represent the same ray, while opposite-parity vertices are orthogonal. Odd cycles and closures that make a nonzero target edge orthogonal are impossible. Exact closure leaves 26 assignments, 18 ordered closure patterns, and nine patterns up to $A \leftrightarrow B$.

For each remaining pattern, assign a formal dot product to each pair of distinct parity components. If vertices i, j have component signs $\sigma_i, \sigma_j \in \{\pm 1\}$, then

$$q_{ij}^X = \frac{1 + \sigma_i \sigma_j x^X_{C(i)C(j)}}{2}. \tag{20}$$

Substitution in the 15 nonzero equations (18) produces a polynomial system over $\mathbb{Q}(\sqrt{7})$. Exact Gröbner reduction gives the unit ideal for eight of the nine exchange classes. Appendix B lists all classes and deletion-minimal contradictory sets; the accompanying checker prints the factored equations. Because a physical Bloch solution would also solve these polynomial equations, a unit ideal is already a conclusive rejection. No positivity relaxation is used in this step.

The sole algebraic survivor, up to exchange of the two factors, has the local closure data

$$\begin{aligned} Z_A &= \{15, 16, 57, 67\}, & O_A &= \{17, 56\}, \\ Z_B &= \{36, 56\}, & O_B &= \{35\}, \end{aligned} \tag{21}$$

where O_X denotes pairs forced to have local fidelity one. Reduction by the surviving transition ideal gives

$$q_{23}^A = q_{34}^A = 1. \tag{22}$$

Unit Bloch vectors with dot product one coincide. Hence rays $2, 3, 4$ in factor A coincide and $q_{24}^A = 1$ as well. The target equations then force in factor B

$$q_{1k}^B = \frac{3}{4} \quad (k = 2, 3, 4), \qquad q_{k\ell}^B = \frac{1}{4} \quad (2 \le k < \ell \le 4). \tag{23}$$

The Bloch Gram matrix of these four local rays would therefore be

$$G = \begin{pmatrix} 1 & 1/2 & 1/2 & 1/2 \\ 1/2 & 1 & -1/2 & -1/2 \\ 1/2 & -1/2 & 1 & -1/2 \\ 1/2 & -1/2 & -1/2 & 1 \end{pmatrix}. \tag{24}$$

But

$$\det G = -\frac{27}{16}, \qquad \operatorname{spec}(G) = \left\{ -\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{3}{2} \right\}. \tag{25}$$

Every real Gram matrix is positive semidefinite, so this survivor is impossible. The factor-exchanged class has the same obstruction.

Lemma 2 (computer-assisted product obstruction). No seven two-qubit product rays reproduce the 21 target transition probabilities.

The argument above proves the lemma, with the finite enumeration and exact ideal calculations supplied as reproducible algebraic certificates. The only non-ideal step for the survivor is the elementary unit-vector implication $\boldsymbol{r} \cdot \boldsymbol{s} = 1 \Rightarrow \boldsymbol{r} = \boldsymbol{s}$, followed by the exact spectrum (25).

5 Strictness and physical scope

Theorem 3. For the states, weights, witness normalization, two-qubit separable preparations, and independent binary effects of Section 2,

$$Q_{\mathrm{SEP}} < \frac{3}{4}. \tag{26}$$

Proof. The two-qubit separable-state set and each effect interval $\{E : 0 \preceq E \preceq I_4\}$ are compact. Their finite Cartesian product is compact, and (8) is continuous, so Q_{SEP} is attained. Equation (15) gives $Q_{\mathrm{SEP}} \leq 3/4$. If equality held, Proposition 1 would produce product rays satisfying (11), in contradiction with Lemma 2. Therefore the inequality is strict. □

Together with the feasible value reported in Ref. [3], the result changes the known interval to

$$0.74990679 \leq Q_{\mathrm{SEP}} < 0.75, \tag{27}$$

where the lower endpoint remains a numerical see-saw value. The strict upper endpoint means that the ideal target correlations lie outside the separable correlation set. Under the assumed four-dimensional message and the prepare-and-measure causal model, the fixed witness therefore has a genuine AES-certification gap for the seven-state extended ensemble.

Several boundaries matter. The result is not device independent; the four-dimensional message assumption is essential. It concerns the extended seven-state ensemble rather than directly certifying the original four-state subset with this witness. It also neither determines Q_{SEP} nor proves that the see-saw point in (27) is globally optimal. Most importantly for experiment, compactness supplies no computable modulus: this work gives no explicit number $U < 3/4$ with certified decimal value and hence no finite noise tolerance.

6 Methods and reproducibility

The target states and weights were independently entered in coordinate and tensor-product representations before the present calculation. Both give unit-normalized rays, positive weights summing to one, the frame identity (6), legal pairwise Helstrom projectors, and the exact value (10).

The product obstruction is implemented in a deterministic Python script using SymPy [4]. It first compares all 21 embedded target transitions against the independently generated machine-readable target input. It then enumerates the 3^6 zero-edge assignments, performs parity closure, constructs the exact systems (20), and computes Gröbner bases over the algebraic extension $\mathbb{Q}(\sqrt{7})$. Rejected systems contain 1 in the ideal; the script also deletes equations until it obtains the cores summarized in Appendix B. The surviving system is reduced exactly to (22)–(23), and the determinant and characteristic polynomial of (24) are evaluated over the rationals. No floating-point comparison or fitted tolerance enters the proof. The code, input, exact output, environment specification, and run instructions accompany the manuscript.

7 Conclusion

The seven-state prepare-and-measure witness cannot be saturated by separable two-qubit preparations. The central mechanism is rigid: equality would copy every target transition probability into a product-ray factorization, while the target's six zero transitions reduce all such factorizations to finitely many local orthogonality patterns. Exact algebra eliminates eight patterns; the last violates positivity of a four-vector Bloch Gram matrix. This proves the existential separation $Q_{\mathrm{SEP}} < 3/4$ without assuming a Wigner property.

The next quantitative task is to make the obstruction stable. A bound that turns a small witness deficit into approximate transition matching, combined with a robust version of the zero-graph and negative-Gram arguments, could produce a rational or interval-certified $U < 3/4$. Such a bound, or a tight moment/SOS dual certificate, is required for an operational experimental threshold and for deciding whether the reported see-saw value is optimal.

Statements and declarations

Data and code availability. All target data used in the proof are displayed in this article. The exact checker, machine-readable input, output certificate, dependency versions, and regeneration instructions are supplied as ancillary files with the manuscript.

Generative-AI disclosure. OpenAI Codex using the GPT-5.6 Sol model assisted with idea development, symbolic-algebra code, exact calculations, literature searching, verification, and manuscript drafting. The AI system is not an author. The corresponding author will review the complete scientific and editorial record and remains responsible for any version that is submitted.

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A Target transition probabilities

For completeness, define

$$u = \frac{7 - 2\sqrt{7}}{14}, \qquad v = \frac{14 - 5\sqrt{7}}{42}. \tag{28}$$

Direct evaluation of (2)–(4) gives

$$(t_{ij})_{i,j=1}^{7} = \begin{pmatrix} 1 & 1/2 & 1/2 & 1/2 & 0 & 0 & u \\ 1/2 & 1 & 1/4 & 1/4 & 1/12 & 1/4 & v \\ 1/2 & 1/4 & 1 & 1/4 & 1/3 & 0 & v \\ 1/2 & 1/4 & 1/4 & 1 & 1/12 & 1/4 & v \\ 0 & 1/12 & 1/3 & 1/12 & 1 & 0 & 0 \\ 0 & 1/4 & 0 & 1/4 & 0 & 1 & 0 \\ u & v & v & v & 0 & 0 & 1 \end{pmatrix}. \tag{29}$$

Every off-diagonal entry is strictly smaller than one. The six vanishing entries in its upper triangle are exactly those in (19).

B Finite enumeration certificate

Table 1 records the nine closure classes produced by the exact parity algorithm. For each factor, Z_X contains pairs forced orthogonal and O_X contains pairs forced identical. The rejected-row entry gives a deletion-minimal set of target-pair equations whose exact Gröbner basis is [1]. The accompanying output lists the corresponding factored polynomials, so each rejection can be replayed without a numerical threshold.

Table 1: Closure classes up to exchange of local factors. Pair labels are one-based and unordered.

Class	A: Z_A/O_A	B: Z_B/O_B	Exact outcome
	$Z = \{15, 16, 57, 67\}$	$Z = \{36, 56\}$	
1	$O = \{17, 56\}$ $Z = \{15, 16, 57, 67\}$	$O = \{35\}$ $Z = \{36, 56, 67\}$	Algebraic survivor; rejected by (24).
2	$O = \{17, 56\}$ $Z = \{15, 36, 57\}$	$O = \{35, 37, 57\}$ $Z = \{16, 56, 67\}$	Unit ideal; core $\{45, 46, 47\}$.
3	$O = \{17\}$ $Z = \{15, 56, 57\}$	$O = \{15, 17, 57\}$ $Z = \{16, 36, 67\}$	Unit ideal; core $\{17\}$.
4	$O = \{16, 17, 67\}$ $Z = \{15, 16, 57, 67\}$	$O = \{13, 17, 37\}$ $Z = \{16, 36, 56\}$	Unit ideal; core $\{17\}$.
5	$O = \{17, 56\}$ $Z = \{15, 57\}$	$O = \{13, 15, 35\}$ $Z = \{16, 36, 56, 67\}$	Unit ideal; core $\{17, 35, 37\}$.
6	$O = \{17\}$ $Z = \{15, 16, 57, 67\}$	$O = \{13, 15, 17, 35, 37, 57\}$ $Z = \{16, 36, 56, 67\}$	Unit ideal; core $\{35, 37\}$.
7	$O = \{17, 56\}$ $Z = \{15, 56, 57\}$	$O = \{13, 15, 17, 35, 37, 57\}$ $Z = \{16, 36, 56, 67\}$	Unit ideal; core $\{45, 46, 47\}$.
8	$O = \{16, 17, 67\}$ $Z = \{15, 36, 57\}$	$O = \{13, 15, 17, 35, 37, 57\}$ $Z = \{16, 36, 56, 67\}$	Unit ideal; core $\{45, 46, 47\}$.
9	$O = \{17\}$	$O = \{13, 15, 17, 35, 37, 57\}$	Unit ideal; core $\{35, 37\}$.

The counts provide additional completeness checks: 26 of the 729 assignments survive local closure; these produce 18 ordered patterns; every ordered pattern is one of the nine rows or its factor exchange. Thus a product-ray solution cannot lie outside the table.

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